

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 1 – C Code Output Prediction**

|                  |                                                                                                               |               |                                              |
|------------------|---------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------|
| Course Code:     | CSA0305                                                                                                       | Course Title: | Data Structures                              |
| CO Assessed:     | CO1 – Imitation (BL3, BL4)                                                                                    | Assessment:   | Assessment Tool 1 – C Code Output Prediction |
| Weightage:       | 35% of CIA                                                                                                    | Total Marks:  | 100                                          |
| CO1 Statement:   | Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. |               |                                              |
| Student Name:    | PRITHAM G                                                                                                     | Reg. No.:     | 192572014                                    |
| Section / Batch: | CSE                                                                                                           | Date:         | 10/07/2026                                   |

**Code of Conduct**

*I PRITHAM G (192572014) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: PRITHAM G

**Instructions**

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20\*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic                                                                   | Focus                                      | Q Nos  | Marks     |
|-----------------------------------------------------------------------------------|--------------------------------------------|--------|-----------|
| Linked data structure                                                             | Basic data structure                       | 1 - 2  | 20        |
| Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure | Classifications of Linear data structure   | 3 - 4  | 20        |
| Search Data Structure                                                             | Linear Search and Binary Search            | 5 -6   | 20        |
| Recursion, Performance analysis                                                   | Recursion                                  | 7 - 8  | 20        |
| Array Data Structures                                                             | Implementations Using Array data structure | 9 - 10 | 20        |
| GRAND TOTAL:                                                                      |                                            |        | 100 Marks |



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# Web Browser Navigation Using Stack (ADT)

**CSA0305 – DATA STRUCTURES**

GUIDE BY

Dr. UMA

PRIYADARSINI

PRESENT BY

PRITHAM G

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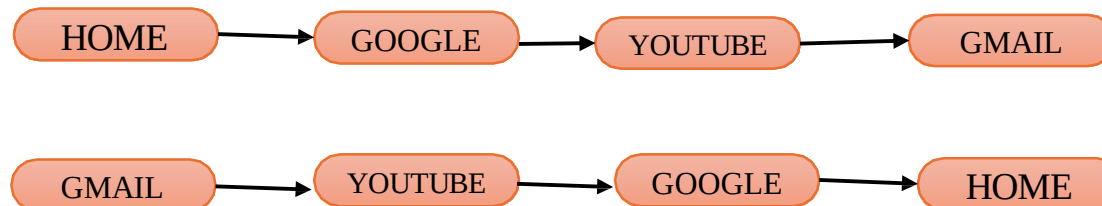
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## INTRODUCTION

What is a Stack?

- Stack is a linear data structure.
- It follows the LIFO (Last In, First Out) principle.
- The last visited webpage is the first one returned when the Back button is pressed.
- Operations used:
  - ✓ Push – Store newly visited webpage.
  - ✓ Pop – Remove current webpage while going back.
  - ✓ Peek – Display current webpage.





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## **Architecture / Algorithm**

### **Working Process**

- User opens a webpage.
- Browser pushes the page into the stack.
- Every new page is pushed on top.
- When Back button is clicked, the top page is popped.
- Browser displays the previous page



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## **Advantages & Performance**

### **Stack is Suitable**

- Fast insertion and deletion.
- Simple implementation.
- Efficient memory usage.
- Provides quick Back navigation.
- Widely used in Chrome, Firefox, Edge, and other browsers



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## CONCLUSION

- Stack is the most suitable ADT for browser Back navigation.
- LIFO perfectly matches browser history behavior.
- Push and Pop operations take **constant time ( $O(1)$ )**, ensuring fast performance.
- Stack improves browsing efficiency and provides a smooth user experience.



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THANK YOU